TRINITY

A Visual Guide to Adversarial Evaluation

Forge the metal, test the metal, remember what the metal cost.

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SECTION 01

The Problem Trinity Solves

Every few months a new AI model comes out and scores higher on benchmarks. For anyone trying to *evaluate* AI capability, that creates a treadmill.

Benchmarks go stale

A test hard in 2023 may be trivial by 2026. The score moves; the skill does not.

Benchmarks leak

Once public, models are trained on it. The test stops measuring anything real.

Benchmarks are gameable

If you know how a test is graded, you optimize for the grader instead of the skill.

The core insight. Difficulty is *measured*, never *claimed*. Write it, execute it against strong models, have someone sign the result, and watch it expire as models improve.

Trinity is the machinery that makes this loop reliable, auditable, and resistant to cheating. It is not just a benchmark generator - it is a **governance system** for benchmark generation.

The treadmill, visually

Step	What happens
1	Model improves - a new SOTA is released.
2	Benchmark softens - the tests now pass easily.
3	Write new tests - assumed hard, never measured.
4	Public leaks - models train on it - hard becomes easy.
5	Loop resets with no guarantee anything was ever truly hard.

SECTION 02

The Three Instruments

Trinity has exactly three core agents. Learn these three and you understand 80% of the system.

MEMORY

ENGRAM

Remembers across tasks, expires stale levers, directs the other two.

Workbench .memory/
Report DIRECTIVE.md

WRITER

FORGE

Authors and proves tasks in batches of thirty slots plus one anchor.

Workbench .seed/
Report EDICT.md

INSPECTOR

CRUCIBLE

Audits the project against instrumented evidence and places clean bundles.

Workbench .audit/
Report VERDICT.md

What each one actually does

- **FORGE** reads human instructions (requirements/) and examples (touchstones/), writes 30 ordinary tasks + 1 anchor per batch, and *seals* each one. It never places tasks - that is CRUCIBLE's job.
- **CRUCIBLE** checks every sealed task for bugs, loopholes, ambiguity, and shortcuts. It runs 12 audit lanes, files findings as GitHub bugs, and places clean bundles into samples/, delivery/, or stock/. It **never sees the difficulty targets**.
- **ENGRAM** keeps the ledger of which task styles have gone soft, tracks which levers still work, publishes the two projections, folds in pilot results, and runs genesis to bootstrap a new project.

Analogy. FORGE is the exam writer. CRUCIBLE is the quality inspector. ENGRAM is the coach who watches the game, remembers what works, and tells the players where to aim - but never plays.

SECTION 03

The Information Barrier

The most important design decision in Trinity. **FORGE and CRUCIBLE never talk to each other.** That is not a bug - it is the whole point.

From	To	What flows
ENGRAM	FORGE	FORGE_VIEW - direction + difficulty targets + levers
ENGRAM	CRUCIBLE	CRUCIBLE_VIEW - standards only, hardness contract withheld
FORGE	CRUCIBLE	**nothing - blocked by construction**

Why: reward hacking

If you are writing exam questions and you can see exactly what the grader will accept, you stop writing questions that test real understanding. You start writing questions that satisfy the grader. This is **reward hacking** - optimizing for the measurement instead of the thing being measured.

In AI terms: if the writer agent knows what the inspector accepts, it learns to write *for the inspector* rather than *for the difficulty goal*. The system collapses into mutual back-patting.

How the barrier is enforced

- **Doctrine.** All three contracts state the rule and follow it.
- **Tooling check.** The parent gate scans every evidence file under .audit/ and refuses any line that cites .memory/hardness.yaml or HARDNESS.md.
- **Submodule boundaries.** .memory/, .seed/, and .audit/ are separate git submodules with separate remotes. The hardness contract sits in a history a parent-only clone never carries.

> **Why it matters.** The barrier is what makes the difficulty claim honest. Without it, the inspector drifts toward whatever the writer produces, and the whole system becomes self-referential.

SECTION 04

The Workflow

Trinity runs in **cycles**, one batch at a time. Four phases, each failing closed on the one before it.

#	Phase	What happens
1	Research (ENGRAM)	Genesis builds the hardness contract before any task exists. An unpublished memory directs nothing.
2	Engineering (FORGE + CRUCIBLE)	FORGE generates the harness/. CRUCIBLE audits it. A pilot whose fidelity coverage was never demonstrated caps at HOLD:SUPPRESSED_MEASUREMENT.
3	Anchoring	The first 3-5 tasks go through written, audited, piloted, signed, folded. Anchors are calibration, not difficulty evidence.
4	Operations	The batch cycle runs to quota: 30 + 1 anchor, sealed, audited, placed. /maestro repeats.

The cycle is one batch wide. FORGE fills staging/ with 30 ordinary bundles + at most one anchor and seals them. CRUCIBLE audits every sealed bundle and places the clean ones - FORGE never places a bundle. The only lawful stops are the named human gates, a terminal disposition, or a blocking gap.

Anchoring in plain words

Anchoring is the calibration pass Trinity runs before mass production. An anchor teaches what this project's levers cost in time. That learning shapes design without upgrading a tier or discharging the pilot obligation. Until three anchors stand ANCHORED across two or more axes, every ordinary batch records budget-provisional or budget-unanchored and caps at HOLD:PILOT_REQUIRED.

SECTION 05

A Batch Lifecycle

Every batch: **30 ordinary tasks + 1 anchor**, sealed, audited, and placed across three tiers with ceilings.

Stage	Who	Output
Staging	FORGE	staging/ - 30 ordinary bundles + at most 1 anchor, all sealed
Audit	CRUCIBLE	Every sealed bundle audited; findings filed as GitHub bugs
Place	CRUCIBLE + pipeline.py	Clean bundles routed to samples/, delivery/, stock/

The three lanes

samples/

Public marketing lane. First 30 bundles in seal order. Fixed by contract.

delivery/

Private corpus. Overflow up to the delivery ceiling (default 10,000).

stock/

Second private tier. Overflow past delivery up to the stock ceiling (default 10,000).

Raising and lowering

- **Raised delivery ceiling** - promotes the oldest of stock/ into delivery/.
- **Lowered delivery ceiling** - demotes the newest of delivery/ into stock/.
- **Past the stock ceiling** - a bundle holds as PIPELINE_STOCK_FULL.

Where ceilings come from (precedence)

- A --delivery or --stock value on the door.
- A ceiling line in requirements/lanes.yaml.
- A line in the hand-edited parent lanes.yaml.
- The default - 10,000.

SECTION 06

Trust and Security

Six layers of defence, each failing closed.

#	Layer	What it guarantees
1	Humans own the inputs	requirements/, touchstones/, playbooks/ are read-only to every agent. Gate refuses agent commits touching them.
2	Evidence is tamper-evident	Every bundle has a content hash. Every human turn lands beside a hash chain. Every commit is signed.
3	Gates fail closed	A gate is a named human stop, approved by writing a SHA-256 digest. No signature = no progress.
4	Sentinel remembers intent	Every intent-class refusal appends to hash chains. A repeat raises an alert capping the gate at BLOCK until cleared.
5	Rank on signed evidence	Nanashi to ronin to akira to tantric. Merit promotes; strikes demote. A tantric co-signs every final SHIP.
6	Release integrity is external	Governance installs release-gate.yaml only in a separate publisher repo, never the candidate.

Dispositions

SHIP
Release. Bare, or with :INFERRED / :CONDITIONAL.

HOLD
Wait. Bare, or with :PILOT_REQUIRED, :SUPPRESSED_MEASUREMENT, :PAROLED.

BLOCK
Refuse. Bare, or with :INVALID_PILOT.

The sentinel remembers intent, never assumes it. A finding code's *shape* is a prior, never intent. A first contact of an omission- or state-shaped code that an accepted lesson receipt answers charges no strike.

SECTION 07

Speak Trinity

A **closed vocabulary** - each term has exactly one meaning, the same across all three instruments.

Term	Meaning
Contract	The Markdown work order you paste into a coding agent. The program *is* the contract; the agent is the runtime.
Instrument	One of the three core agents: ENGRAM, FORGE, CRUCIBLE.
Workbench	The directory an instrument owns: .memory/, .seed/, or .audit/. Exactly three.
Harness	The harness/ benchmark extension FORGE generates. Never an instrument's directory.
Roster root	One of nine submodules against its own remote: trinity/, .trinity/, .memory/, .seed/, .audit/, samples/, delivery/, stock/, harness/.
Root report	DIRECTIVE.md, EDICT.md, or VERDICT.md at the project root.
Lever	A reusable technique for making a task harder.
Tier	A difficulty band: Baseline, Hard, or Frontier-defeat.
Pilot	An external signed run of a frozen task against a frozen solver registry. The only proof a task is hard.
Task lane	Machine-decided routing: samples/, delivery/, or stock/.
Anchor task	One of the first 3-5 tasks FORGE authors, piloted first.
Disposition	The grade a gate emits: SHIP, HOLD, or BLOCK, bare or with a substate.
Gate	A named human stop, approved by writing a SHA-256 digest.

One source, every surface. Doctrine that several contracts state identically lives once under doctrine/. The three contracts include it; compose.py renders it into GAUNTLET and MAESTRO inline.

SECTION 08

Getting Started

Two paths. Pick the one that matches where you are.

PATH A

Existing project

Vendor the trinity/ submodule into a project that already has requirements/ and touchstones/.

```
git submodule add -b main <trinity remote> trinity
git config -f .gitmodules submodule.trinity.update merge
uv run --project ./trinity --no-dev
./trinity/tools/gate.py install ./
```

Then run /engram, /forge operations, /crucible operations, one at a time. Sign each gate as it stops.

PATH B

New project

Run ENGRAM genesis to birth a whole knowledge repository from scratch.

```
uv run --project ./.memory engram genesis --target
./projects/<project>
```

Genesis creates the hardness contract, the 11-file doc spine, eight roster roots, governance spine, and mascot art.

Your first day (one time)

- Register: identity.py register ./ --login <your login>.
- Sign the printed request, then run identity.py accept.
- Install doors: gate.py install ./ --signing-key <key>, then one signed commit.
- Run the gate with --check --no-record first - nothing is recorded, so a mistake gives no strike.
- Read NEXT STEPS, correct each finding, then do one real run.

What this makes you: a **nanashi** - you may author FORGE slots. CRUCIBLE needs **ronin**, granted by a tantric. ENGRAM needs a tantric from tantric.keys. Until you register, the gate treats you as an external ronin: your runs hold and your pushes block.

When a gate refuses you

Read the code before you act. Do not run the gate again hoping it clears - each recorded sabotage-class finding is a strike. Do not edit trinity/, the ledgers, or a sealed bundle. The Refusal bestiary in LEARNING.md explains the codes you meet first.

SECTION 09

Philosophy and Reference

The six non-negotiables

- **Difficulty is measured, never claimed.** Only an external signed pilot counts.
- **Separation prevents reward hacking.** FORGE and CRUCIBLE never talk. ENGRAM is the only bridge.
- **Everything fails closed.** Missing anything means HOLD or BLOCK.
- **Humans own the inputs.** Agents cannot edit the instructions they follow.
- **Evidence is tamper-evident.** Content hashes, signed commits, hash chains.
- **Nothing lasts forever.** Tasks expire as models improve.

The two meta-instruments

GAUNTLET

Out-of-band gatekeeper. Pasted only where authorized. Tries to break the artifact a gate binds before discharging. Keeps no workbench of its own.

MAESTRO

Conductor. Sequences ENGRAM, FORGE, and CRUCIBLE in fixed order and consolidates pending gates. Conducts and never performs. Keeps no state.

What to read next

Read it for	Document
Onboarding	LEARNING.md - ARCHITECTURE.md - STORY.md
Work orders	ENGRAM.md - FORGE.md - CRUCIBLE.md - GAUNTLET.md - MAESTRO.md
People and authority	ACTORS.md - SIGNOFF.md - docs/trust-root.md
Honesty	DEFERRED.md
History	CHANGELOG.md - RELEASE.md - ORION.md
Decisions	docs/adr/ - 95 records
Tooling	templates/README.md - templates/SENTINEL.md

Read DEFERRED.md before you evaluate any claim here. It registers every obligation the contracts state that nothing enforces - including the external signed pilot that keeps FORGE at HOLD:PILOT_REQUIRED. Trinity would rather tell you what it cannot do yet than pretend it can.

TRINITY

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